Troy Mayer

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Open to Summer 2027 internships in robot software and controls

EDUCATION

Purdue University, West Lafayette, IN Expected May 2028
B.S. Mechanical Engineering and B.S. Applied Statistics, Minor in Economics GPA: 3.32/4.00

SKILLS

- **Languages:** C++, Python, MATLAB, SQL
- **Robotics:** state estimation (Kalman filtering), feedback control, QP force control, rigid-body kinematics, MuJoCo, Eigen, FreeRTOS
- **Tools:** Linux, Git, CMake, GoogleTest, GitHub Actions CI, Flask, pandas, PyTorch, KiCad, LabVIEW, CATIA, Creo

ROBOTICS EXPERIENCE AND PROJECTS

Boston Dynamics Jul 2026 – Present
Sensors and Perception Co-Op (Atlas) Waltham, MA

- Mapping Atlas impacts onto the robot mesh with **floating-base forward kinematics** from joint encoders, and estimating impact energy from operational-space effective mass and a fitted impact velocity.
- Designed a delta-v impact-severity metric that stays accurate when sensors miss an impact's peak, and characterized its filtering and noise floor with synthetic and null-control tests.
- Building an impact forensics pipeline and dashboard (Python, Flask, SQLite) over ~28k impacts from Atlas test logs, used by engineers during testing.

Legged Robot State Estimation and Control, *personal project, code* Aug 2026 – Oct 2026

- Built a **contact-aided error-state Kalman filter in C++** for a simulated Unitree Go1, fusing IMU and leg kinematics to 0.013 m/s velocity error, against 1.38 m/s for the IMU alone.
- Verified it with GoogleTest (finite-difference Jacobian checks, parity with a Python reference) and an observability analysis of what the sensors can and can't see.
- Stress-tested it against foot slip and contact latency, found that tilt is only weakly observable on straight walks, and fixed the tilt and latency failures.
- Closed the loop in Python simulation, with the walking controller running on the estimate without a fall; also added an MIT Cheetah 3-style QP balance controller.

Purdue ASME Biomechanics, Active Orthosis Apr 2024 – May 2026
Electronics Team Lead West Lafayette, IN

- Designed a **1 kHz PI motor velocity controller in C++** for an ESP32-S3, tuned from a motor model with feed-forward and anti-windup, and verified it in co-simulation with a model of the motor and forearm.
- Wrote the FreeRTOS firmware around it, with fault monitors, a hardware watchdog, and a kill switch, tested in CI with GoogleTest.
- Cut EMG-to-motion latency from 103 to 43 ms in simulation on real recorded EMG by reworking the signal processing and loop tuning.
- Led the 5-engineer electronics team and redesigned the EMG board schematic in KiCad, raising cross-day gesture decoding from 31% to 87% in emulation.

Asteroid Hopping Robot, *independent research with Purdue University and UC3M (Madrid)* Oct 2026 – Present

- Building a three-legged hopping robot and a landing test rig, and designing its push-off, landing, and mid-air attitude control with MuJoCo simulation.

ADDITIONAL EXPERIENCE

Caterpillar (via Purdue Data Mine), *Machine Learning Researcher* Jan 2026 – May 2026

- Built LSTM and XGBoost forecasts of daily paint consumption with Monte Carlo confidence intervals, contributing to a 4.7% reduction in material usage.

Chortos Lab, Purdue, *Undergraduate Researcher* Aug 2025 – May 2026

- Wrote a Python computer-vision tracker and LabVIEW/MATLAB data acquisition to measure gel samples, and used them to formulate gels with over 50% more force output.

Emerson, Discrete Automation (ASCO), *Mechanical and Manufacturing Engineer Intern* Jun 2025 – Aug 2025

- Designed and CNC-machined a spring canister for an industrial rotary valve in Creo with GD&T drawings, and ran UL 429 validation testing on solenoid valves.